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Pre-aeration of food waste to augment acidogenic process at higher organic load: Valorizing biohydrogen, volatile fatty acids and biohythane

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HIGHLIGHTS

- Aeration of feedstock enhanced biohydrogen and VFA production.
- Hydrolysate formed during aeration of feedstock enhanced microbial metabolites during fermentation.
- Oxygen present in the air suppressed methanogens and resulted in the formation of higher VFA from food waste.
- Production of H_2 and CH_2 in a single stage bioprocess showed formation of biohythane.

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ABSTRACT

Application of pre-aeration (AS) to waste prior to feeding was evaluated on acidogenic process in a semi-pilot scale biosystem for the production of biobased products (biohydrogen, volatile fatty acids (VFA) and biohythane) from food waste. Oxygen assisted in pre-hydrolysis of waste along with the suppression of methanogenic activity resulting in enhanced acidogenic product formation. AS operation resulted in 97% improvement in hydrogen conversion efficiency (HCE) and 10% more VFA production than the control. Increasing the organic load (OL) of food waste in association with AS application improved the productivity. The application of AS also influenced concentration and composition of fatty acid. Highest fraction of acetic (5.3 g/l), butyric (0.7 g/l) and propionic acid (0.84 g/l) was achieved at higher OL (100 g COD/l) with good degree of acidification (DOA). AS strategy showed positive influence on biofuel (biohydrogen and biohythane) production along with the biosynthesis of short chain fatty acids functioning as a low-cost pretreatment strategy in a single stage bioprocess.

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1. Introduction

Food waste (FW) is a rich source of carbohydrates and proteins which can function as a potential feedstock to valorize spectrum of biobased products including fuels (Sen et al., 2016; Lin et al., 2013; Reddy et al., 2011). The quantities of FW have been increasing rapidly and therefore needs to essentially establish a sustainable and effective remediation strategy (Girotto et al., 2015). Most of the FW routes to landfills making the disposal practice unsustainable, terminating the option for resource recovery and leads to release of green house gases (Thia et al., 2016). Chemical and/or thermal methods conventionally deployed to treat FW are low in economic feasibility and manifest environmental impact with gaseous emissions. Alternatively, biological processes can utilize waste as a feed stock for production of biofuels and biobased products (Venkata Mohan et al., 2016; Sen et al., 2016; Lin et al., 2013).

Recently, acidogenic process using FW was reported for the production of biohydrogen (H_2), volatile fatty acids (VFA), biohythane (CH_4:H_2 = 80:20), etc. along with remediation (Sarkar et al., 2016). In general, acidogenic process is very dynamic and depends on many biotic and abiotic factors. Among them, the nature of biocatalyst and feed-stock (wastewater) composition and load plays a critical role in overall process efficiency (Goud and Venkata Mohan, 2012; Goud et al., 2014; Pasupuleti et al., 2014). Hydrogenotrophic methanogens utilize H_2 and carbon dioxide to produce methane (CH_4) leading to the formation of bio-hythane, which have good heat of combustion, improved flammability and burning speed (Pasupuleti and Venkata Mohan, 2015). However, methanogens impends yield of biobased products in particularly H_2 and VFA (Venkata Mohan et al., 2008; Venkata Mohan, 2009). To certain extent, an increment in organic load (OL) positively affects the acidogenesis process (Goud and Venkata Mohan, 2012). However,

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feeding higher OL becomes load-shock to the biosystem. FW is generally associated with very high OL, which will have detrimental effect on the acidogenesis due to excessive production of fatty acids that manifest to suppress system buffering capacity leading to acidic redox microenvironment effecting bio-catalytic function.

In this study, an attempt was made to overcome the persistent limitation encountered with methanogens reactivation during the higher OL operation in a semi-pilot scale bioreactor, pre-aeration strategy (AS) to FW was applied to improve overall acidogenesis process efficiency. Molecular oxygen is known to induce inhibitory effects on the methanogens which are strict anaerobes (Lim and Wang, 2013). This approach also accelerates the break-down (hydrolysis) of organic substances into soluble mono and oligomers (monomer sugars, amino acids and fatty acids) by hydrolytic enzymes released by aerobes which enhances hydrolysis of waste and renders more energy recovery from waste (Pedizzi et al., 2016; Botheju et al., 2009). However, continuous air sparging reduces accumulation of VFA, resulting in a low drop of pH during the process, thus improving the efficiency at start-up of the bioprocess (Ahn et al., 2014; Cossu et al., 2016; Peces et al., 2016). AS is a cost effective and simple approach compared to other pretreatment techniques to regulate selective production of bio-based production.

2. Material and methods

2.1. Food waste

Composite food waste (FW) was collected from institute canteen and stored at 4 °C. The food waste prior to use was masticated using electrical blender and later filtered through a stainless-steel sieve to remove coarse materials. Oil present in the waste was separated using a density gradient separating funnel with a settling time of 36 h. The oil free filtrate obtained was used as a substrate after adjusting the various organic loads (OL) based on COD viz., 60, 70, 80, 90, and 100 g COD/l by tap water at pH 6. Food waste was aerated using a regulating air pump for 60 min with ~400 l of O_2 calculated using a gas flow meter (INSCIN; wet type) prior to feeding the reactors. A non-aerated food waste was used for the control operations to determine the variations induced by the AS. The food waste was analyzed for pH, COD, nitrogen, protein and solids (APHA, 1998). Raw food waste showed a high organic load (3.15 ± 0.3 kg COD/l) with a suitable biodegradable fraction (BOD/COD-0.72 ± 0.2).

2.2. Biocatalyst

Anaerobic culture from the laboratory scale bioreactors was used as parent inoculum after subjecting to heat-shock treatment (HST) at 80–90 °C for a period of 2 h to inhibit the methanogens (Venkata Mohan et al., 2008). The resulting parent inoculum was enriched in design synthetic wastewater (DSW; NH_4Cl-0.5 g/l, KH_2PO_4-0.25 g/l, K_2HPO_4-0.25 g/l, $MgCl_2$-0.3 g/l, $CoCl_2$-25 mg/l, $ZnCl_2$-11.5 mg/l, $CuCl_2$-10.5 mg/l, $CaCl_2$-5 mg/l, $MnCl_2$-15 mg/l, $NiSO_4$-16 mg/l, $FeCl_3$-25 mg/l) under anaerobic conditions at pH 6 for 48 h.

2.3. Experimental

Semi-pilot scale anaerobic hybrid biofilm reactor designed with a L/D ratio of 6 (length × diameter; 120 × 20 cm) and coir pith (void fraction 0.18) as fixed bed packing material to support the biofilm formation was used for the experiments (Pasupuleti et al., 2014). The reactor has a working/total volume of 20/34 l and a gas holding capacity of 4.5 l (head space). The bioreactor was operated in a fed-batch mode with an up-flow velocity of 0.50 m/day (14 l/day) at mesophilic temperature (30 ± 2 °C) and anaerobic microenvironment. A peristaltic pump (EYELA 2000m) was used to feed the food waste and to regulate the recirculation during operations. Initially the bioreactor was operated with 60 g COD/l after preparing the feed by adjusting the COD (control). To evaluate the influence of air sparging, the bioreactor was then loaded with air-sparged feed stock at 60 g COD/l. Thereafter, the bioreactor was operated with air sparged feed at higher organic load from 70–100 g COD/l. All the cycles were operated at a hydraulic retention time (HRT) of 54 h.

2.4. Analysis

The performance of process was monitored through periodic sampling and analysis. Biohydrogen was estimated using a microprocessor based pre-calibrated H_2 sensor (ATMI GmbH Inc., Germany). Biogas composition was analyzed using gas chromatography (GC; NUCON 5765) with thermal conductivity detector (TCD) with 1/8″ × 2 m Heysep Q column and employing argon as carrier gas. The injector and detector were maintained at 60 °C each and the oven was operated at 40 °C isothermally. The bioprocess monitoring was done by assessing the parameters viz. chemical oxygen demand (COD-closed refluxing titrimetric method), volatile fatty acids (VFA) and pH (APHA, 1998). VFA (acetate, propionate, butyrate, valarate) composition was estimated by using HPLC (Shimadzu LC10A) employing UV–Vis detector (210 nm) and C18 reverse phase column (250 × 4.6 mm diameter; 5 μm particle size, flow-rate: 0.5 ml/h; wave length: 210 nm). Mobile phase of (40% acetonitrile in 1 mN H_2SO_4; pH, 2.5–3.0) and 20 μl of filtered sample (0.22 μm porosity) injection was used. Standard mix acids (SUPELCO, USA) were used for quantitative estimation of the VFA produced. Alcohols were analyzed using HPLC (Shimadzu LC20A) employing RI detector (RID20A; Shimadzu) and rezex monosaccharide (Phenomenex) column by injecting filtered sample of 20 μl. For elution, a flow rate of 0.5 ml/min was used in an isocratic method with water as the mobile phase and the column temperature was maintained constant at 80 °C in a column oven (RA20A).

2.5. Data analysis

Hydrogen conversion efficiency (HCE) is defined as the percentage of feedstock converted to H_2. Theoretically, 1 kg of COD (food waste) can produce 468.83 l of H_2 (based on acetate pathway of dark fermentation process) (Pasupuleti et al., 2014). The hydrogen conversion efficiency of the food waste in the specified reactor was calculated using Eq. (1).

$$\textit{Hydrogen conversion efficiency}\ (HCE, \%) = \frac{CHP * 10,000}{OL * COD_R * V * THY} \quad (1)$$

where, CHP is cumulative hydrogen production (L), OL is the organic loading (g COD/l), COD_r is the substrate removal efficiency (%), V is the substrate feeding volume to the reactor (l) and the theoretical hydrogen yield (THY; 0.468 l/g COD_r). Degree of Acidification (DOA) represents the extent of acidification achieved due to the production of carboxylic acids in relation to substrate (as COD) degradation (Eq. (2)) (Sarkar et al., 2016).

$$\textit{Degree of Acidification}\ (DOA, \%) = \frac{S_f}{S_i} \times 100 \quad (2)$$

where S_i represent initial substrate concentration (COD; mg/l) and S_f is net VFA concentration expressed as theoretical equivalents

factor of COD (in mg/l, H_{Ac}, 1.066; H_{Pr}, 1.512; H_{Bu}, 1.816). Buffering capacity (β) was estimated using auto-titration with the Eq. (3).

$$\beta = \frac{C}{V_s \times m} \tag{3}$$

where C is the concentration of acid or base (mol), V_s is the volume of sample (ml), m is the slope of tangent on curve (Sarkar et al., 2016). Production and consumption rates of carboxylic acids were calculated based on the following equations (Eqs. (4) and (5)) (Sarkar et al., 2017).

$$\text{Production rate of carboxylic acids}(PR_{Ca}) = (VFA_{max} - VFA_{int})/T_{prod} \tag{4}$$

$$\text{Consumption rate of carboxylic acids}(CR_{Ca}) = (VFA_{drop} - VFA_{max})/T_{prod} \tag{5}$$

where VFA_{max} represents maximum VFA concentration (g/l/h), VFA_{int} is initial VFA concentration (g/l/h), VFA_{drop} denotes drop/consumption in VFA concentration due to its consumption (g/l/h), T_{prod} is production time in hours and T_{drop} represents concentration dropping/consumption time (hours). The positive and negative values explained the rate of production and consumption of carboxylic acids respectively. Production was calculated from the initial hour to the time of next sampling while the consumption was calculated from the point of degradation to end of the cycle for each acid individually. All the experiments were operated in triplicates the data provided is consolidation of three individually operated repeats.

3. Result and discussion

3.1. Pre-aeration of waste: acidogenic product profile

3.1.1. Biohydrogen

Pre-aeration of waste prior to feeding showed a marked influence on the acidogenic metabolites viz., H_2, VFA and CH_4. Fig. 1a depicts the H_2 production profile operated for 72 days (36 cycles). The first six cycles (control) were operated at 60 g COD/l without pre-aerated feedstock as control. The next 30 cycles were operated at different OL (60 to 100 g COD/l) with pre-aerated feedstock. A pre-aerated feedstock (diffusing of 2200 l of air) showed positive influence on H_2 production and yielded a 1.79-fold increase in cumulative hydrogen production (CHP) compared to the control operation (75.8–136 l). Higher OL become load-shock where accumulation of soluble metabolites acidifies the system microenvironment which deteriorates the acidogenesis process. Pasupuleti et al. in their study observed excess accumulation of soluble metabolites at higher OL operation (60 g COD/l) which resulted in the cease of H_2 production (Pasupuleti et al., 2014).

Improved H_2 production by air sparging can be attributed to the suppression of the methanogenic activity and the rapid hydrolysis of carbohydrates (almost 20 times faster) present in the waste (Lay et al., 2003). The higher OL showed an improved H_2 evolution compared to the control operations and the lower OL with air sparged food waste. Overall increase in the CHP was noticed at all OL viz. 60 g COD/l, 70 g COD/l, 80 g COD/l, 90 g COD/l and 100 g COD/l operated with aerated feedstock with 1.11, 1.32, 1.61, 1.79 and 1.67 folds increment respectively. CHP increment in a range of 105–136 l was recorded with the maximum CHP observed at 90 g COD/l. The percentage of H_2 in biogas showed a gradual increase from 12% (6 h) to 30% (30 h) followed by a drop from 24% (36 h) to 14% (48 h) (Fig. 1b).

Biohydrogen production from food waste at optimal controlled fermentation (pH, nature of inoculum) depends on the nature of substrate used (Venkata Mohan et al., 2012; Goud et al., 2014).

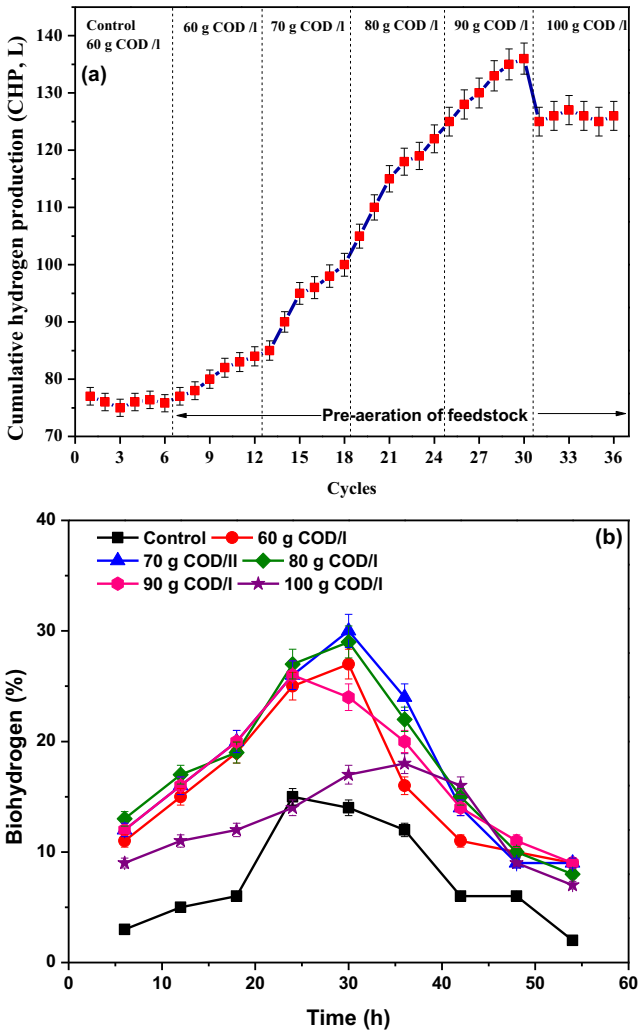


Fig. 1. (a) Influence of air sparging (AS) on cumulative hydrogen production (CHP) during acidogenic fermentation of composite food waste compared to control (b) H_2 evolved in the headspace of the bioreactor during fermentation of food waste.

The AS of food waste solubilizes the complex particulate matter like carbohydrates of food waste by the catalytic activity of extracellular enzymes secreted by hydrolytic microorganisms during aeration (Lim and Wang, 2013). Solubilized simpler sugar molecules are a perfect substrate for the anaerobic bacteria to produce H_2 which was evident from the gas composition analysis (Chu et al., 2012; Hallenbeck, 2009). Aeration also enhances the hydrolysis of carbohydrates and proteins in the waste as well as the activities of cellulase and protease hydrolytic enzymes (Johansen and Bakke, 2006; Rafieenia et al., 2017).

3.1.1.1. Waste remediation. A distinct variation in waste remediation (COD removal) pattern was observed. Application of pretreatment showed lower substrate degradation (COD_r: 58%) with ASP operation compared to the control (78%), (Fig 2). Higher substrate degradation can be attributed to the activity of methanogens which links to CH_4 production. Despite of relatively low substrate degradation, conversion to bio-based products (H_2, VFA, etc.) was higher than the control operation. At 70 g COD/l, maximum COD_r recorded was 61% which gradually improved to 69/76% (80/90 g COD/l). Overall the highest substrate degradation (83%) was noticed with 100 g COD/l operation.

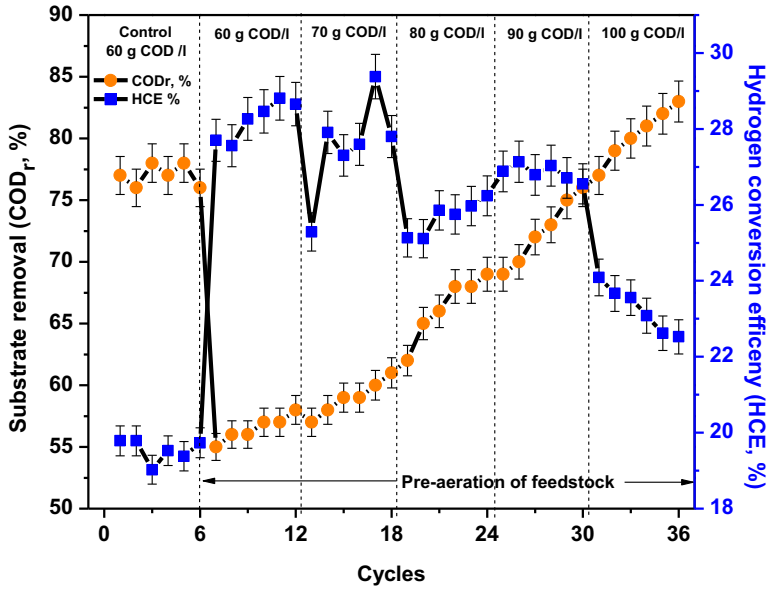


Fig. 2. Influence of air sparging strategy (AS) and higher organic load on substrate degradation (CODr) and hydrogen conversion efficiency (HCE) compared with control operation.

3.1.1.2. Hydrogen conversion efficiency (HCE). Theoretically, 1 kg of COD (wastewater) can produce 468.8 l of H_2 (based on acetate-dark fermentation pathway). Application of pre-aeration strategy showed marked improved in HCE compared to control operation (control: 19.7%; AS: 28.8%). Increasing OL (70–90 g COD/l) documented marginal improvement in HCE (29.3–26.2%). At 70 g COD/l operation, highest HCE (29.4%) was achieved due to higher acidogenic and lower methanogenic activity (Fig 2). Operation at 90 g COD/l documented a HCE of 27.3% (Cycle 26), whereas at 80 g COD/l operation the HCE was in the range of 25–26%. It is apparent from the experimental data that despite higher orgnic load, the AS can possibly inhibit methanogens which aid in the evolution of good H_2 resulting in good HCE. HCE recorded at 100 g COD/l was comparatively lower than all other OL. The ranking of HCE with higher OL compared to control is 29.3% (70) > 27.1% (90) > 26.2% (80) > 19.7% (Control).

3.1.2. Biohythane

The gas fraction of CH_4 gradually increased with time at different OL (Fig 3a). At 6 h, CH_4 content during all the OL operation was in the range of 7–14%, which gradually increased to 28–35% by the end of the cycle operation. Increasing OL resulted in higher CH_4 at end of the cycle operation which can be attributed to the activation of methanogenic activity due to presence of H_2 in head space as well as VFA. Highest CH_4 of 35% was noticed at 100 g COD/l. Higher substrate degradation also attributed to the persistence of methanogens at the later stages of cycle operation (Goud and Venkata Mohan, 2012; Pasupuleti et al., 2014). Higher CH_4 in the reactor is the result of VFA consumption which was known to be better precursors for methanogenesis (Basso et al., 2009). Methanogens play a synergetic role, where products of acidogenic and acetogenic reaction serve as the substrate for methanogens (S1 and 2).

High fraction of CH_4 along with low concentration of H_2 (between 42 and 54 h), favored biohythane formation. A mixture of hydrogen and methane in an appropriate concentration is termed as biohythane, which can function as alternative renewable fuels (Pasupuleti and Venkata Mohan, 2015; Sarkar and Venkata Mohan, 2016). Fig. 3b depicts the flexible H_2 content by volume ($H_2/(H_2 + CH_4)$) (CO_2 was excluded) ranging between 0.20 and 0.60 with respect to time and organic load, which falls in the suggested range (0.1–0.25) at the end of each cycle operation. The increase in organic load influenced the production of biohythane with respect to operation time. From the experimental data, cumulative biohythane production found varying based on organic load from 128.7 l (control; 60 g COD/l), 144 l (60 g COD/l), 156 l (70 g COD/l), 156 l (80 g COD/l), 159 l (90 g COD/l), and 163 l (100 g COD/l) (Sfig. 1).

3.1.3. Volatile fatty acids

The feedstock pre-aeration application had documented positive influence on the carboxylic acids (VFA) production in correlation with the H_2 production. Figure 4a depicts the total VFA production with respect to fermentation time at varying OLs. Application of pre-aeration to waste improved VFA production by 10% from 4596 mg COD/l (control) to 5087 mg/l at 60 g COD/l operation. Improved yield in the fatty acid can be attributed to the resulting suppression of methanogenic activity during the feeding stage. Increasing the carbon load showed marked increment in VFA production (70 g COD/l, 5869 mg/l; 80 g COD/l, 6155 mg/l; 90 g COD/l, 6466 mg/l; 100 g COD/l, 6754 mg/l) at 36 h of the cycle operation. Exceeding the cycle period beyond 34 h resulted in consumption of VFA in all the cases due to fact that the H_2 and CO_2 get utilized along with VFA to form CH_4. Methanogenic activity leads to consumption of VFA which was specifically viewed between 36 and 54 h (control, 4596–3587 mg/l; AS operation, 5087–4516 mg/l). As most of the methane is derived from the methyl carbon of acetate and the carboxyl carbon gets converted to carbon dioxide (Zeikus, 1980; Dahiya et al., 2015). Short chain fatty acids production has been well regulated with the AS strategy even at high OL operation.

3.1.3.1. Fatty acid profile and acidification. Application of pre-aeration also showed marked influence on fatty acid profile. Acetic acid (H_{Ac}), Butyric acid (H_{Bu}) and Propionic acid (H_{Pr}) were the main short chain VFA produced by the catalytic activity of acidogenic bacteria in this study. VFA composition and its dynamics with time showed some specific observations (Fig. 4c–h). Quantitatively, H_{Ac} was major fraction of all the fatty acids produced, followed by butyric and/or propionic acids (in couple of cases) with varied concentrations in conjugation with time. The propionic acid

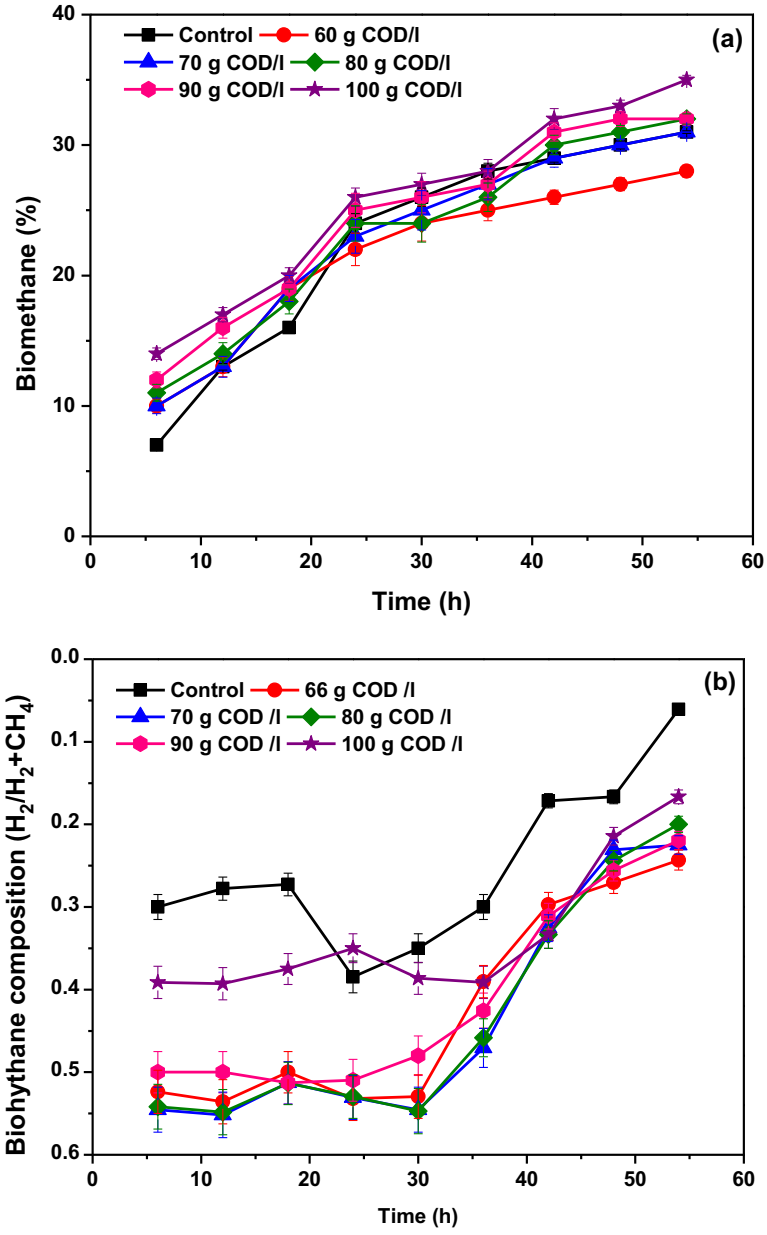


Fig. 3. (a) Cumulative biohythane production recorded at higher OL in comparison with control (b) Composition of biohythane with respect to fermentation time at different organic load.

production was higher than butyric acid in the case of control (60 g COD/l) and higher OL operation (100 g COD/l). H_{Ac} production peak was consistently observed at 40 h in all the cases and dropped afterwards. H_{Bu} and H_{Pr} production showed a gradual increment up to the end of cycle operation, but varied with OL. Higher H_{Ac} followed by H_{Bu} correlated well with the H_2 production. AS effectively improved H_{Ac} (4100 mg/l; 40 h) and H_{Bu} (710 mg/l; 40 h) than control (H_{Ac}: 3566: 40 h) and (H_{Bu}: 315 mg/l; 48 h). Increasing OL (70–100 g COD/L) improved H_{Ac} and H_{Bu} concentrations. H_{Pr} production was higher with un-aerated feedstock (control 855 mg/l) (control) and 100 g COD/l (844 mg/l) of operation. However small amount of H_{Pr} was also produced during AS operation with varied concentration of 533 mg/l, 366 mg/l, 300 mg/l and 296 mg/l at 90 g COD/l, 80 g COD/l, 70 g COD/l and 60 g COD/l respectively. Order of VFA species distributed in control and 100 g COD/l is $H_{Ac} > H_{Pr} > H_{Bu}$ whereas in 60, 70, 80 and 90 g COD/l the order was $H_{Ac} > H_{Bu} > H_{Pr}$.

Effect of pre-aeration was quite visible on DOA (Fig 4b). Near about 5% improvement has been noticed in DOA after application of pre-treatment (from 19.51% to 24.84%). Higher concentrations of acetate, propionate and butyrate at increasing OL improved the DOA. The highest DOA (29.7%) was noticed at 70 g COD/l. On the contrary, despite high OL, 100 g COD/l showed least DOA of 11.24%. Relatively good DOA of 20% and 16.7% was observed with 70 and 80 g COD/l of loading respectively.

Owing to acidification, the reactor showed a detrimental trend in pH (below 5) which stimulated solventogenesis that led to the production of ethanol in very low concentrations (Sarkar et al., 2017). The ethanol presence was observed higher with 70 g COD/l (0.1 g/l; 36 h) operation in comparison to 90 g COD/l (0.085 g/l; 36 h), 80 g COD/l (0.078 g/l; 36 h), 100 g COD/l (0.07 g/l; 36 h), 60 g COD/l (0.05 g/l; 36 h) and control (0.04 g/l; 36 h). The variations in ethanol concentration are marginal.

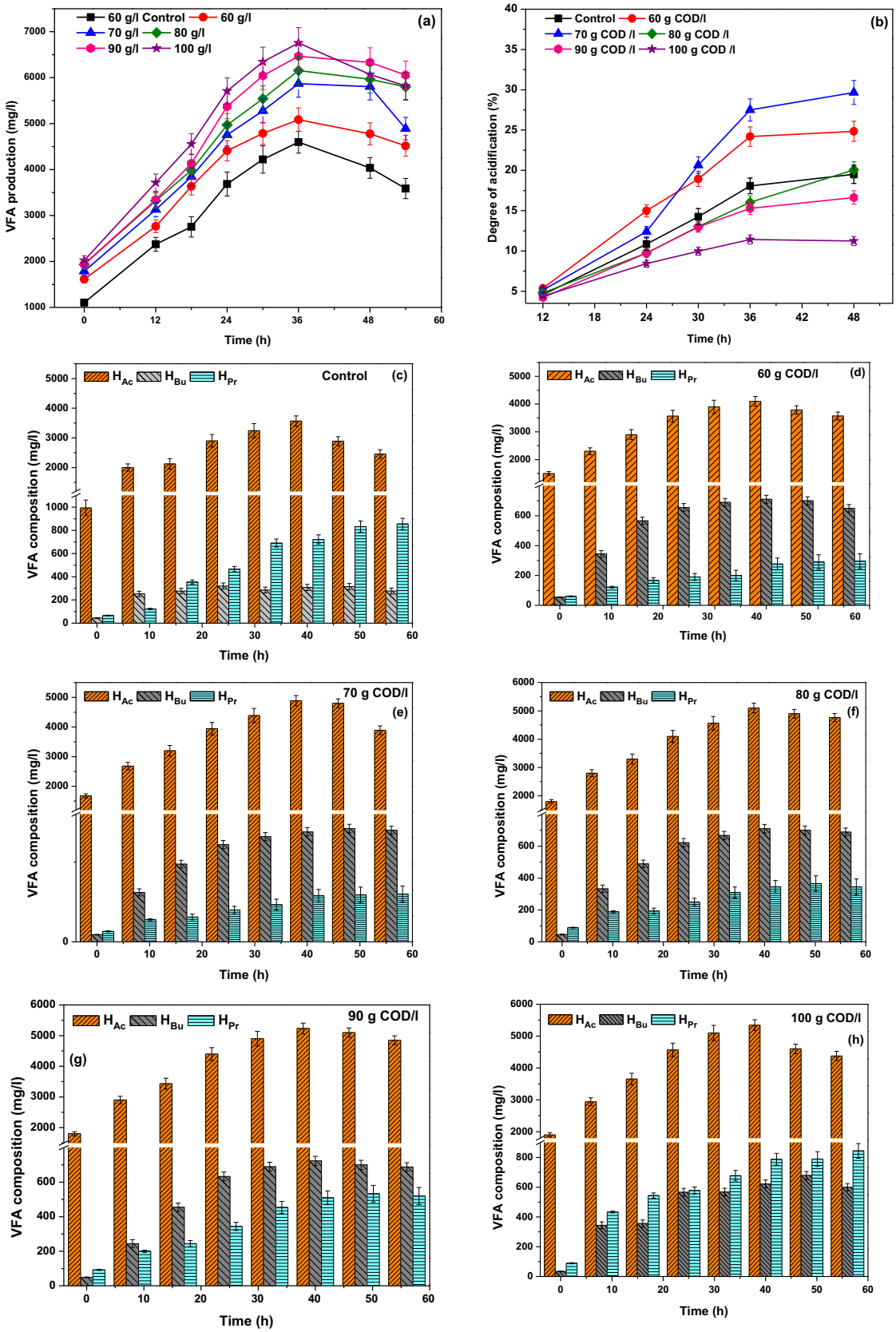


Fig. 4. Application of AS on total Volatile fatty acid production (TVFA), (b) and degree of acidification (DOA) and VFA composition (c–h) with respect to time.

3.1.3.2. VFA kinetics: production and consumption rate. The acidogenic process is always dynamic which gets influenced by the fermentation metabolites particularly the quantity and composition of the fatty/carboxylic acids (Sarkar et al., 2017). The pattern of acids produced and consumed varies with fermentation time and operation conditions (Fig. 5a–f). Pre-treatment showed a positive

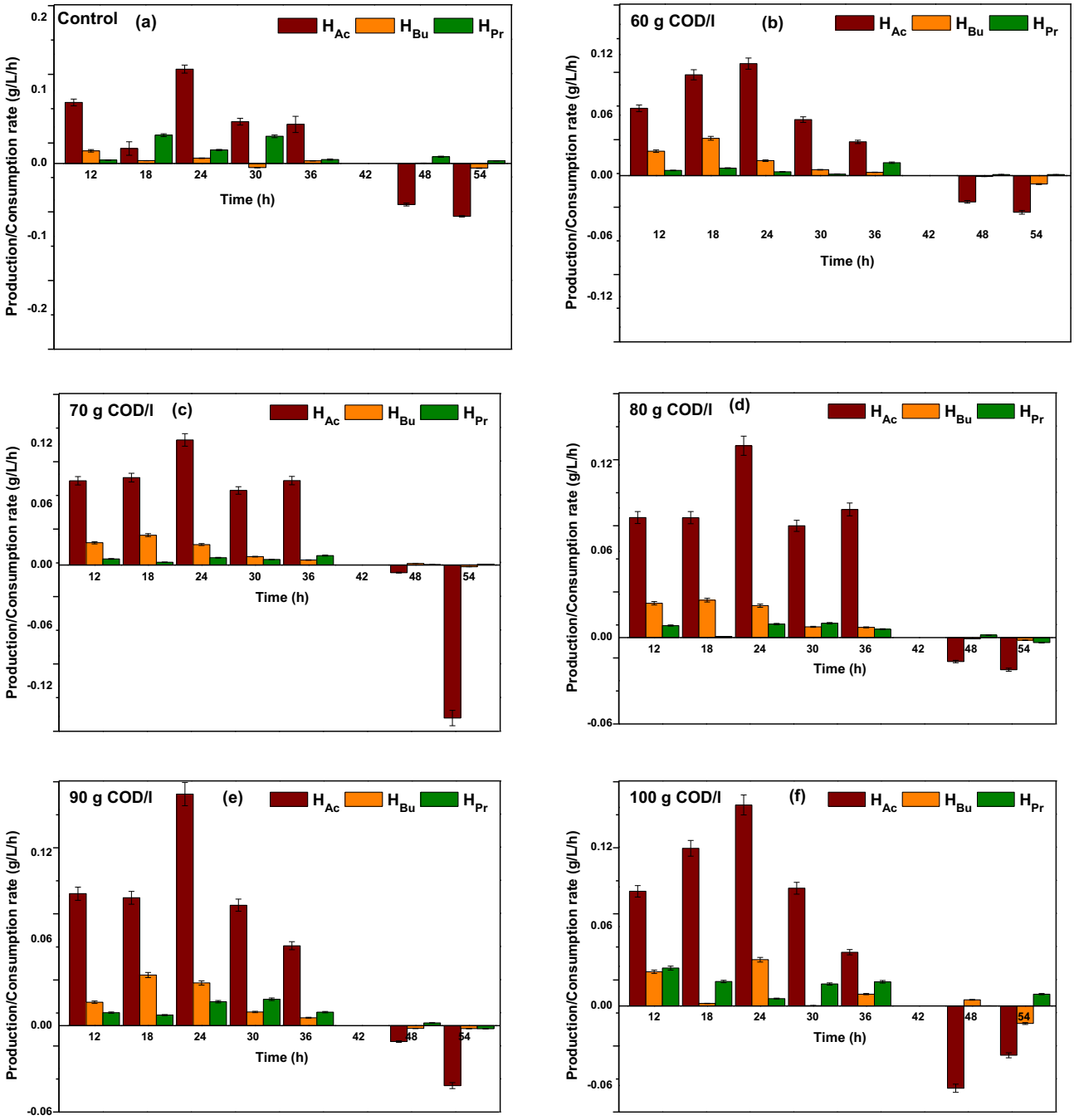


Fig. 5. Consumption and production pattern of volatile fatty acids during fermentative H_2 production at variable organic load.

influence on production and consumption rate of fatty acids depicting higher H_{Ac} (0.111 g/l/h) and H_{Bu} (0.037 g/l/h) production rate compared to control (H_{Ac}: 0.13 g/l/h; H_{Bu}: 0.018 g/l/h). However, control showed higher H_{Pr} (0.039 g/l/h) production than AS (0.013 g/l/h). Control operation documented high H_{Ac} consumption rate (0.009 g/l/h) than AS (0.001 g/l/h). Consumption of H_{Ac} was the indication for the formation of methane. Usually, most of the methane is derived from the methyl carbon of acetate and the carboxyl carbon which then is converted to carbon dioxide (Zeikus, 1980; Sarkar et al., 2016; Dahiya et al., 2015) (Eq. (7)). (S3).

Highest H_{Ac} production rate of 0.16 g/l/h was noticed at 90 g COD/l followed by 0.15 g/l/h, 0.13 g/l/h, and 0.12 g/l/h at 100 g COD/l, 80 g COD/l and 70 g COD/l respectively. Higher H_{Ac} production at elevated OL is also correlated with H_2 production in the initial phase of the fermentation. However, consumption of H_{Ac} varied with the fermentation time. At 70, 80 and 90 g COD/l, higher H_{Pr} production of 0.0094 g/l/h, 0.01 g/l/h and 0.019 g/l/h was documented respectively. Consumption of H_{Pr} was also seen at different time interval of fermentation of −0.0035 g/l/h (54 h; 80 g COD/l) and −0.0021 g/l/h (90 g COD/l; 54 h). Proton reducing acetogens converts H_2 and CO_2 with propionic acid and further form CH_4 by methanogens (Angenent et al., 2004) (S4). Production rate of H_{Bu} (0.036 g/l/h) was higher at 90 g COD/l followed by 0.035 g/l/h, 0.026 g/l/h and 0.029 g/l/h at 100 g COD/l, 80 g COD/l and 70 g COD/l respectively. (S5).

3.2. Redox microenvironment

System pH plays a critical role in metabolic pathways of acidogenic bacteria towards specific product synthesis (Wang et al.,

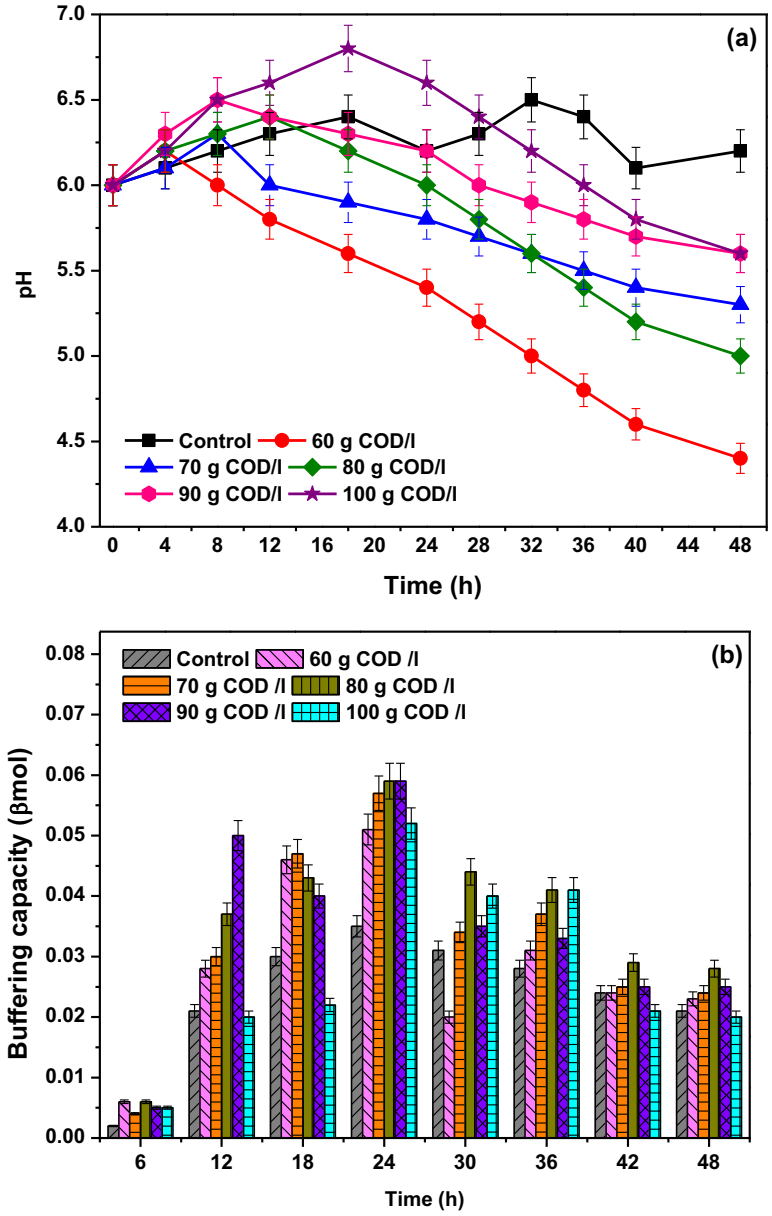


Fig. 6. (a) Change in pH with respect to time; (b) Buffering capacity analyzed at different time interval.

2014). The studied biosystem was operated under uncontrolled initial pH 6. During the course of operation marginal pH increment (pH 6–6.4) was observed in control operation whereas, in AS and higher OL operation increment in pH was observed in only in initial hours of the cycle operation which later decreased towards the end of the cycle [60 g COD/l: 6–6.4 (8 h) to pH 4.4 (48 h); 70 g COD/l: pH 6.3 (8 h) to pH 5.3(48 h); 80 g COD/l: pH 6.4 (12 h) to pH 5 (48 h); 90 g COD/l; pH 6.5 (8 h) to pH 5.6 (48 h)] (Fig 6a). Change of pH in a particular range during acidogenesis is an indicator of production, accumulation, and consumption of specific acid (Dahiya et al., 2015; Sarkar et al., 2016). The pH range 5–5.5 thus observed during AS operation at 18 h resulted in the dominance of H_{Bu}.

Fatty acid production and composition influences the system's buffering capacity (BC) (Sarkar et al., 2016). Buffering condition keeps the pH of solution constant by taking up protons that are released during bacterial substrate metabolism (Venkata Mohan et al., 2017). BC gradually increased from 6 to 24 h followed by decrement till the end of the cycle (Fig 6b). Comparatively AS exhibited higher BC (0.051 βmol) with higher H_2 (27%) than control (0.035 βmol; H_2:15%) at 24 h. Acetoclastic and hydrogenotrophic methanogenesis enable methane production during the process of acidogenesis with mixed microbial consortia (Hao et al., 2008). The evolution of CH_4 occurs with the significant role of syntrophic acetate oxidation even at acidic redox condition (pH 5) to produce bicarbonate (Hattori, 2008) (S6). This bicarbonate triggers the system buffering condition which was correlated with the BC pattern observed in this study during AS and higher OL operation compared to control.

Depending on the concentration and availability of acetate in the bioreactor, acetoclastic methanogenesis may shift to hydrogenotrophic methanogenesis even at lower redox conditions (pH 4). Hydrogenotrophic methanogenesis involves the conversion of CO_2/HCO_3- to CH_4 in the presence of H_2 (Kotsyurbenko et al., 2007) (S7 and 8).

Higher CH_4 evolution between 30 and 54 h resulted in the formation of biohythane may be attributed to the action of

hydrogenotrophic methanogenesis which promoted in the bioconversion of H_2 to CH_4 even at acidic microenvironment (pH 4.5–5.5).

4. Conclusions

Application of low-cost pre-aeration to food waste before feeding showed significant influence on the acidogenic process and metabolite formation. Molecular oxygen present in air not only suppressed the methanogens but it also assisted in the hydrolysis of complex food waste to simpler molecules. It is apparent from the experimental data that pre-aeration strategy could inhibit methanogens and assisted in good H_2 production and other acidogenic metabolites. Relatively good biohydrogen production was also observed at higher organic load. AS documented the feasibility and potential as pretreatment option for bioconversion of waste to biobased products with simultaneous remediation in a waste biorefinery mode.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.biortech.2017.05.053.

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